

DS 2500 *Fall* 2025 Lab Exercise 09

Deadline : 11/10/2025 11:59 PM EST

Late Deadline: 11/24/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Setup Instructions

- Use the following code to begin this lab

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split

NUMERICAL_FEATURES = ['length', 'diameter', 'height', 'whole_weight',
                      'shucked_weight', 'viscera_weight', 'shell_weight',
                      'rings', 'age']

TARGET = 'sex'
```

- We will use the same data from HW03 for this lab. To load data into a Pandas DataFrame, follow the [Use Pandas to Load and Preprocess the Data](#) section in the HW03 description. If you have already completed this part in HW03, you can simply reuse your existing dataframe. For this lab instruction, let's assume it is called `df` -- feel free to replace `df` with your own variable name.

Problem 1 (50 pts)

- Convert the `sex` feature from `df`.

- The `sex` feature originally has three classes: M, F, and I. Filter out all rows where `sex == 'I'` to make this lab into a binary classification task.
- Then run the following code.

```
df['sex'] = df['sex'].map({'F': 1, 'M': 0})
```

- Note that this modification applies only for Lab 09.
- Use `train_test_split()` to split the dataframe `df` into 2 dataframes.
 - `test_size` must be set to `0.2`.
 - `random_state` must be set to `0`.
- The resulting dataframes should be named as `df_tr` and `df_te`.

Problem 2 (50 pts)

- Use `LogisticRegression()` to instantiate a model.
- Train the model using `NUMERICAL_FEATURES` from the training set.

Problem 3 (50 pts)

- Find the most important feature.
 - You can either write code to find it (recommended) or just look at them and find one manually.
- Store the name of the feature into the variable `most_important_feature`.

Problem 4 (50 pts)

- Make predictions using the test set.
- Generate the summary of the model performance using `classification_report()`.
- From that report, compare two metrics for predicting the positive class: precision and recall, and store the metric with the higher value in the variable `problem4_answer`.
 - For example, if precision was higher, then

```
problem4_answer = 'precision'
```